Predicting Demand and Detecting Anomalies in Urban Bike-Share Systems

A Data Mining Project Proposal

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Abstract

This project proposes a data mining pipeline for analyzing urban bike-share systems using real-world trip-log data. The goal is twofold: (1) to predict short-term station-level demand so that operators can proactively rebalance bicycles across stations, and (2) to detect anomalous trips that may indicate theft, vandalism, system malfunction, or data-quality errors. The project will combine data preprocessing and warehousing, clustering, classification, frequent pattern mining, and outlier detection — the core techniques covered in this course — into a single end-to-end case study. Publicly available datasets (Citi Bike NYC trip histories, paired with public weather data) will be used to ensure reproducibility. Success will be measured by predictive accuracy relative to simple historical baselines and by the precision of flagged anomalies against known operational incidents.

1. Introduction and Problem Statement

Bike-share systems such as Citi Bike (New York), Divvy (Chicago), and Santander Cycles (London) have become a core part of urban transportation, serving tens of millions of trips per year. A persistent operational challenge for these systems is the *demand-supply imbalance*: stations near business districts empty out during the morning commute while stations near residential areas overflow, and vice versa in the evening. Operators must predict where bikes will be needed and physically truck bicycles between stations ("rebalancing") — an expensive and logistically complex task. At the same time, bike-share operators must monitor the system for anomalous behavior: trips with implausible

duration or distance, which may signal theft, mechanical failure, or GPS/sensor errors in the underlying data feed.

This project asks: **can we use historical trip, weather, and calendar data to (a) predict near-term station demand well enough to support rebalancing decisions, and (b) automatically flag anomalous trips worth investigating?** This is a practical, data-rich problem that touches on time-series/tabular classification, unsupervised clustering of spatial-usage patterns, frequent sequential patterns in commuter behavior, and outlier detection — making it a natural capstone for the techniques covered in this course.

2. Related Work

Bike-share analytics has been studied extensively in the data mining and ubiquitous computing literature:

- Froehlich, Neumann, and Oliver (2009) were among the first to analyze large-scale bike-share data (Barcelona's Bicing system), showing that station usage exhibits strong, predictable temporal and spatial regularities that can be exploited for forecasting.
- O'Mahony and Shmoys (2015) studied New York City's Citi Bike system directly, formulating station rebalancing as an optimization problem informed by historical demand patterns, and highlighting the real operational cost of poor demand prediction.
- Li, Zheng, Zhang, and Chen (2015) proposed a hierarchical prediction model (clustering stations into groups before predicting cluster-level demand) to improve accuracy and reduce the variance caused by sparsely-used stations — an approach directly relevant to the clustering component of this proposal.
- Chen et al. (2016) introduced a dynamic, cluster-based method for predicting "over-demand" stations (where supply cannot meet demand) in bike-share systems, explicitly combining clustering and classification, which closely motivates the hybrid approach proposed here.

This project builds on these works by (1) reproducing a simplified version of the clustering + demand-classification pipeline on recent Citi Bike data, and (2) extending it with an explicit outlier-detection and frequent-pattern-mining layer that prior work treats

less thoroughly, in order to practice the full breadth of techniques covered in this course rather than optimizing a single model in isolation.

3. Proposed Work: Data, Tools, and Techniques

3.1 Data

- **Primary dataset:** Citi Bike NYC trip history data (publicly available, monthly CSV files at `citibikenyc.com/system-data`), containing trip start/end time, start/end station ID and coordinates, and user type.
- **Auxiliary data:** Daily/hourly weather data (temperature, precipitation) from NOAA or a public weather API, and a calendar feature set (day-of-week, holiday flag), used to enrich the demand-prediction model.

3.2 Tools

Python (pandas, scikit-learn), `mlxtend` for frequent pattern/association rule mining, `PyOD` for outlier detection baselines, SQLite/Parquet for the local data warehouse, and Jupyter notebooks for exploratory analysis and reporting.

3.3 Tasks and Techniques

Stage	Technique	Purpose
Data preprocessing	Cleaning, missing-value handling, feature engineering	Remove duplicate/invalid trips, derive trip duration/distance, merge weather & calendar features
Data warehousing	Star schema (fact: trips; dimensions: station, time, weather)	Support efficient aggregation queries for modeling and reporting
Clustering	k-means / DBSCAN on hourly station usage vectors	Group stations into usage-profile types (commuter hub, residential, tourist/leisure)
Classification	Random Forest / Gradient Boosting	Predict high- vs. low-demand station-hours using time, weather, and cluster-membership features

Frequent pattern mining	FP-Growth / association rules	Discover common origin-destination trip sequences and station pairings
Outlier detection	Isolation Forest / statistical thresholds (PyOD)	Flag trips with anomalous duration, distance, or route for manual review

4. Evaluation Plan

- **Demand classification:** Accuracy, precision/recall, and F1-score on a held-out time period (e.g., the final two weeks of the dataset), compared against a naive baseline that predicts the historical average demand for the same station-hour-of-week.
- **Clustering quality:** Silhouette score and qualitative inspection (do clusters align with known land-use types near each station, e.g., using station location/borough metadata?).
- **Frequent patterns:** Support/confidence/lift of discovered rules, manually reviewed for plausibility (e.g., do high-confidence station pairs correspond to known commuter corridors?).
- **Outlier detection:** Since ground-truth theft/malfunction labels are not directly available, precision will be approximated via manual review of a random sample of flagged trips, supplemented by any publicly reported system outage dates as a weak validation signal.

Definition of success: The project will be considered successful if the demand classifier outperforms the historical-average baseline by a meaningful margin (target: +10% F1), the clustering produces interpretable, stable station groups, and the outlier-detection stage surfaces a manageable, plausible set of flagged trips (not an overwhelming false-positive rate) in manual review.

5. Timeline

Phase	Duration	Deliverable
1. Data acquisition & preprocessing	Week 1–2	Cleaned dataset, merged weather/calendar features, data warehouse schema built
2. Exploratory analysis & clustering	Week 3	Station usage clusters, exploratory visualizations

3. Classification & frequent pattern mining	Week 4–5	Trained demand classifier, mined association rules
4. Outlier detection & evaluation	Week 6	Anomaly detection results, full evaluation metrics
5. Final report & presentation	Week 7–8	Final written report, presentation slides, code repository

References

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